(and, in a certain sense, incomplete, since anharmonicity can produce an unstable dynamic from a stable linearized problem). Nevertheless such a stability threshold can be found by numerical experiment and it can be verified that this is not too far from the stability threshold that would correspond to a model linearized at equilibrium points.

The only way to increase the stability threshold is by introducing implicitness or by changing in some fundamental way the timestepping framework. We consider some alternatives below.

4.1 Implicit Schemes

By introducing a coordinate transformation, we may reduce the harmonic oscillator system to a pair of decoupled complex scalar ODEs of the form

$$\dot{z} = \pm i\Omega z.$$

Applying a Runge-Kutta method to such a linear system allows direct determination of the stability condition. For example, Euler's method would yield

$$z_{n+1} = (1 \pm ih\Omega)z_n,$$

which exhibits a growing solution $|z_{n+1}| = \left(1 + h^2\Omega^2\right)^{n/2} |z_0|$.

The Implicit Midpoint method which, for the ODE $\dot{z} = f(z)$ takes the form

$$z_{n+1} = z_n + h\frac{f(z_n) + f(z_{n+1})}{2},$$

can be applied to $\dot{z} = i\Omega z$ and yields

$$z_{n+1} = \frac{1 + ih\Omega/2}{1 - ih\Omega/2} z_n.$$

For all real Ω, we find

$$\left|\frac{1 + ih\Omega/2}{1 - ih\Omega/2}\right| = 1.$$

This indicates that the Implicit Midpoint method has no stability threshold for the stepsize when solving a linear system! This remarkable property may inspire some optimism that Implicit Midpoint would allow much larger stepsize in molecular dynamics simulation than does Verlet.

In actual practice, this does not work out quite as expected. To understand why not, we need to recognize that the linear stability condition only describes the